

Fie $R = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in M_2(\mathbb{R}) : \theta \in \mathbb{R} \right\} (= SO(2))$

a) Arătați că R este p.o. în $(M_2(\mathbb{R}), \cdot)$; b) $(R, \cdot)$?

Sol: Fie $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, $B = \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} \in R$

$$BA = AB = \dots = \begin{pmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{pmatrix} \in R$$

b) " " pe R este comutativă

" " are elem neutru $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{pmatrix}$

$$\Rightarrow A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \Rightarrow A^{-1} = \begin{pmatrix} \cos(-\alpha) & -\sin(-\alpha) \\ \sin(-\alpha) & \cos(-\alpha) \end{pmatrix}$$

$GL_n(\mathbb{R})$

$$O(n) = \{ A \in M_n(\mathbb{R}) \text{ inversabilă} : A^{-1} = A^+ \}$$

$$SO(n) = \{ A \in O(n) : \det A = 1 \}$$

Determinanti :

Fie K corp comutativ ($K \in \{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$)

Fie $n \in \mathbb{N}^*$. Reamintim că $(M_n(K), \cdot)$ are elem

$$U(M_n(K)) = \{ A \in M_n(K) : \det A \neq 0 \} =$$

$$= GL_n(K)$$

(= grupul general liniar de grad n peste K)

Def: Fie $A \in M_n(K)$, $A = (a_{ij})$

$$\det A = \sum_{\sigma \in S_n} \varepsilon(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)} \in K$$

Not: $\det A = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$

Obs: $n=2 \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

Proprietăți:

0) $\det A = \det(A^T)$ (A^T e mai not ${}^t A$)

1) Dacă A are 2 lin/col egale $\Rightarrow \det A = 0$

2) Dacă A are o lin/col formată numai din zero $\Rightarrow$
 $\Rightarrow \det A = 0$

3) Dacă A are 2 lin/col proporționale $\Rightarrow \det A = 0$

4) Dacă B se obține din A prin permutarea a 2 lin/col $\Rightarrow$
 $\Rightarrow \det B = -\det A$

5) Dacă B se obține din A prin înmulțirea unei lin/col
cu o constantă $\lambda \in K \Rightarrow \det B = \lambda \det A$

6) Dacă B se obține din A prin înmulțirea unei lin/col
cu $\lambda \in K$ și adăugarea la alte $\Rightarrow \det B = \det A$

7) Fie $A_{ij} \in M_{n-1}(K)$, $i, j \in \{1, \dots, n\}$ not obținute din
A prin eliminarea liniei și coloanei elem a_{ij}

not: $d_{ij} = \det A_{ij}$ ("minorul" lui a_{ij})

$\Delta_{ij} = (-1)^{i+j} d_{ij}$ ("Complementul algebric" al lui a_{ij})

Atunci: $\det A = a_{11}\Delta_{11} + a_{12}\Delta_{12} + \dots + a_{1n}\Delta_{1n}, \forall i = \overline{1, n}$

ex: Calculati

$$a) \begin{vmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{vmatrix} \begin{array}{l} C_2 - C_1 \rightarrow C_1 \\ C_3 - C_1 \rightarrow C_3 \\ C_4 - C_1 \rightarrow C_4 \end{array} \begin{vmatrix} 2 & -1 & -1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} =$$

$$= 1 \cdot (-1)^{2+2} \cdot \begin{vmatrix} 2 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 1 \cdot 1 = 1$$

$$b) \begin{vmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{vmatrix} = 2 \cdot (-1)^{4+4} \cdot \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix} +$$
$$1 \cdot (-1)^{4+3} \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{vmatrix} + 1 \cdot (-1)^{3+4} \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{vmatrix} + (-1) \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{vmatrix} =$$

$$= 2 \cdot (2^3 - 2 - 2) - (2^2 - 1) - (2^2 - 1) =$$

$$= 2(8 - 4) - (4 - 1) - (4 - 1) =$$

$$= 2 \cdot 4 - 3 - 3 =$$

$$\boxed{= 2}$$

$$\begin{array}{l} p1 \ C_4 - 2C_3 \\ p2 \ \text{def. } (-1)^{4+3} \\ p3 \ C_1 - 2C_2 \end{array}$$

$$c) \begin{vmatrix} -1 & a & a & \dots & a & a \\ a & -1 & a & \dots & a & a \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a & a & a & \dots & a & -1 \end{vmatrix} = (\text{ord } n \geq 2)$$

$L_1 + L_2 + \dots + L_n \rightarrow L_n \Leftrightarrow$
 $L_1 + L_2 + \dots + L_n$

$$= \begin{vmatrix} (n-1)a-1 & (n-1)a-1 & \dots & (n-1)a-1 \\ a & -1 & \dots & a \\ \vdots & \vdots & & \vdots \\ a & a & \dots & -1 \end{vmatrix} =$$

$$= [(n-1)a-1] \begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ a & -1 & a & \dots & a \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & \dots & \dots & -1 \end{vmatrix} \begin{matrix} C_2 - C_1 \\ C_3 - C_1 \\ \vdots \\ C_m - C_1 \end{matrix}$$

$$= [(n-1)a-1] \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ a & -1-a & 0 & \dots & 0 \\ a & 0 & -1-a & \dots & 0 \\ a & 0 & 0 & \dots & -1-a \end{vmatrix} =$$

$$= [(n-1)a-1] \cdot \begin{vmatrix} -1-a & 0 & \dots & 0 \\ 0 & -1-a & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 1 & \dots & -1-a \end{vmatrix} =$$

$$= [(n-1)a-1] \cdot (-a-1)^{n-1} \cdot \underset{1}{I_{n-1}} =$$

d) $\begin{vmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_1 \\ x_3 & x_1 & x_2 \end{vmatrix}$, unde $x_1, x_2, x_3 \in \mathbb{C}$ sunt mod: $x^3 - 2x^2 + 2x + 17 \in \mathbb{Q}[x]$

$$\begin{cases} x_1 + x_2 + x_3 = 2 \\ x_1 x_2 + x_1 x_3 + x_2 x_3 = 2 \\ x_1 x_2 x_3 = -17 \end{cases}$$

$$\begin{aligned} & \begin{matrix} L_1 + L_2 + L_3 \\ \hline \end{matrix} \begin{vmatrix} x_1 + x_2 + x_3 & x_1 + x_2 + x_3 & x_1 + x_2 + x_3 \\ x_2 & x_3 & x_1 \\ x_3 & x_1 & x_2 \end{vmatrix} = \end{aligned}$$

$$= (x_1 + x_2 + x_3) \begin{vmatrix} 1 & 1 & 1 \\ x_2 & x_3 & x_1 \\ x_3 & x_1 & x_2 \end{vmatrix} \begin{matrix} C_2 - C_1 \\ \hline \\ C_3 - C_1 \end{matrix}$$

$$= (x_1 + x_2 + x_3) \begin{vmatrix} 1 & 0 & 0 \\ x_2 & x_3 - x_2 & x_1 - x_2 \\ x_3 & x_1 - x_3 & x_2 - x_3 \end{vmatrix} =$$

$$= (x_1 + x_2 + x_3) \begin{vmatrix} x_3 - x_2 & x_1 - x_2 \\ x_1 - x_3 & x_2 - x_3 \end{vmatrix} =$$

$$= (x_1 + x_2 + x_3) \left((x_3 - x_2)(x_2 - x_3) - (x_1 - x_2)(x_1 - x_3) \right) =$$

$$= (x_1 + x_2 + x_3) \left(x_3 x_2 - \cancel{x_3^2} - \cancel{x_2^2} + x_2 x_3 - \left(x_1^2 - x_1 x_3 - x_1 x_2 + \cancel{x_2 x_3} \right) \right) =$$

$$= (x_1 + x_2 + x_3) \left(- (x_1^2 + x_2^2 + x_3^2) - (x_2 x_3 + x_1 x_3 + x_1 x_2) \right) =$$

$$= (x_1 + x_2 + x_3) \left[3(x_1x_2 + x_1x_3 + x_2x_3) - (x_1 + x_2 + x_3)^2 \right] =$$

$$= 2(3 \cdot 2 - 2^3) = 2(6 - 8) = -4$$

e)
$$\begin{vmatrix} x_1 & x_2 & \dots & x_{n-1} & x_n \\ x_2 & x_3 & & x_n & x_1 \\ \vdots & \vdots & & & \\ x_n & x_1 & & x_{n-2} & x_{n-1} \end{vmatrix} = 0, \quad n \geq 2 \text{ unde } x_1, \dots, x_n \in \mathbb{C}$$

sunt rădăcini

$$\rightarrow x^n + a_{n-2}x^{n-2} + \dots + a_1x + a_0 \in \mathbb{C}[x]$$

$$a_{n-1} = 0 \Rightarrow \det = 0$$

deoarece $x_1 + x_2 + \dots + x_n = 0$

f) Rezolvati in $\mathbb{C}$ ecuatia:

a)
$$P(x) = \begin{vmatrix} x & a & a & a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} \begin{matrix} L_1 + L_2 + L_3 + L_4 \\ = 0 \end{matrix}, \quad a \in \mathbb{C}$$

$$\Rightarrow P(x) = \begin{vmatrix} x+3a & x+3a & x+3a & x+3a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} =$$

$$= (x+3a) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} =$$

$$= (x+3a) \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & x-a & 0 & 0 \\ 0 & 0 & x-a & 0 \\ 0 & 0 & 0 & x-a \end{vmatrix} =$$

$$= (x+3a)(x-a)^3 \Rightarrow \begin{cases} x+3a=0 \\ (x-a)^3=0 \end{cases} \Rightarrow \begin{aligned} x_1 &= -3a \\ x_2 &= a \\ x_3 &= x_4 \end{aligned}$$

$$b) \quad P(x) = \begin{vmatrix} x & 0 & -1 & 1 & 0 \\ 1 & x & -1 & 1 & 0 \\ 1 & 0 & x-1 & 0 & 1 \\ 0 & 1 & -1 & x & 1 \\ 0 & 1 & -1 & 0 & 0 \end{vmatrix} \begin{matrix} C_3+C_4 \\ =0 \end{matrix}$$

$$P(x) = \begin{vmatrix} x & 0 & 0 & 1 & 0 \\ 1 & x & 0 & 1 & 0 \\ 1 & 0 & x-1 & 0 & 1 \\ 0 & 1 & x-1 & x & 1 \\ 0 & 1 & -1 & 0 & x \end{vmatrix} \begin{matrix} C_1 - xC_4 \\ = \end{matrix}$$

$$= \begin{vmatrix} 0 & 0 & 0 & 1 & 0 \\ 1-x & x & 0 & 1 & 0 \\ 1 & 0 & x-1 & 0 & 1 \\ -x^2 & 1 & x-1 & x & 1 \\ 0 & 1 & -1 & 0 & x \end{vmatrix} = - \begin{vmatrix} 1-x & x & 0 & 0 \\ 1 & 0 & x-1 & 1 \\ -x^2 & 1 & x-1 & 1 \\ 0 & 1 & -1 & x \end{vmatrix} \begin{matrix} C_1+C_2 \\ = \end{matrix}$$

$$= - \begin{vmatrix} 1 & x & 0 & 0 \\ 1 & 0 & x-1 & 1 \\ 1-x^2 & 1 & x-1 & 1 \\ 1 & 1 & -1 & x \end{vmatrix} \begin{matrix} C_2 - xC_1 \\ = \end{matrix} - \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & -x & x-1 & 1 \\ 1-x^2 & 1+x^3-x & x-1 & 1 \\ 1 & 1-x & -1 & x \end{vmatrix} \begin{matrix} C_1+C_2 \\ = \end{matrix}$$

$$= - \begin{vmatrix} -x & x-1 & 1 \\ 1+x^3-x & x-1 & 1 \\ 1-x & -1 & x \end{vmatrix} \begin{matrix} L_2-L_1 \\ = \end{matrix} - \begin{vmatrix} -x & x-1 & 1 \\ x^3+1 & 0 & 0 \\ 1-x & -1 & x \end{vmatrix} =$$

$$\begin{aligned}
 &= + (1+x^3) \begin{vmatrix} x-1 & 1 \\ -1 & x \end{vmatrix} = (1+x^3) \cancel{(x-1)} + 1 = \\
 &= (1+x^3) (x^2 - x + 1) = \\
 &= (x+1) (x^2 - x + 1)^2
 \end{aligned}$$

$$P(x) = 0 \Rightarrow (x+1) (x^2 - x + 1)^2 = 0$$

$$\Rightarrow \begin{cases} x+1=0 \Rightarrow x_1 = -1 \\ x^2 - x + 1 = 0 \end{cases}$$

$$\Delta = 1 - 4 = -3 < 0 \Rightarrow x_2, x_3, x_4, x_5 \in \mathbb{C}$$

$$\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$$

Ex: For $n \in \mathbb{N}$ $n \geq 2$, $a_1, \dots, a_n \in \mathbb{C}$. Argue that:

$$V(a_1, \dots, a_n) = \begin{vmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \\ a_1^2 & a_2^2 & \dots & a_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} \end{vmatrix} = \prod_{1 \leq i < j \leq n} (a_j - a_i)$$

$$(x_1, y_1)$$

$$(x_2, y_2)$$

$$\vdots$$

$$(x_n, y_n)$$

$$f(x) = a_0 + \dots + a_{n-1}x^{n-1}$$

$$\begin{aligned} f(x_1) &= y_1 \\ &\vdots \end{aligned}$$

$$\begin{cases} a_0 + a_1x_1 + \dots + a_{n-1}x_1^{n-1} = y_1 \\ a_0 + a_1x_2 + \dots + a_{n-1}x_2^{n-1} = y_2 \\ \vdots \\ a_0 + a_1x_n + \dots + a_{n-1}x_n^{n-1} = y_n \end{cases}$$

M_I

(inductive terms)

M_{II}

polynome

not $P(x) = V(a_1, \dots, a_{n-1}, x)$

$$\begin{vmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_1 & & \\ a_1^2 & a_2^2 & & \\ \vdots & \vdots & & \\ a_1^{n-1} & a_2^{n-1} & & \boxed{x^{n-1}} \end{vmatrix}$$

$$P(a_1) = 0, P(a_2) = 0, \dots, P(a_{n-1}) = 0$$

$$\Rightarrow P(x) = c(x-a_1)(x-a_2)\dots(x-a_{n-1})$$

↑
coef bei x^{n-1}

Deswegen: $P(x) = \begin{vmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_1 & & \\ a_1^2 & a_2^2 & & \\ \vdots & \vdots & & \\ a_1^{n-1} & a_2^{n-1} & & \end{vmatrix} x^{n-1} + \dots + V(a_1, \dots, a_{n-1})$

pt $x = a_n \Rightarrow V(a_1, \dots, a_n) = (a_{n-1})(a_{n-2}) \dots (a_n - a_{n-1})$